



Chapter 7: Reporting and Analyzing Inventory

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Inventory Classification

Inventory Classification Depends on Business Model

Service Companies

No Inventory – sell services only

Merchandising Companies (Wholesaler or Retailer)

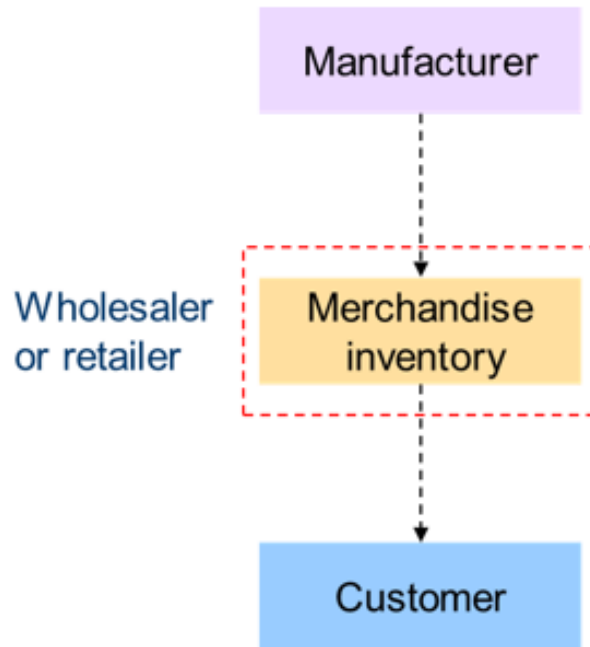
- Merchandize Inventory – Ready for Sale
- Single category

Manufacturing Companies

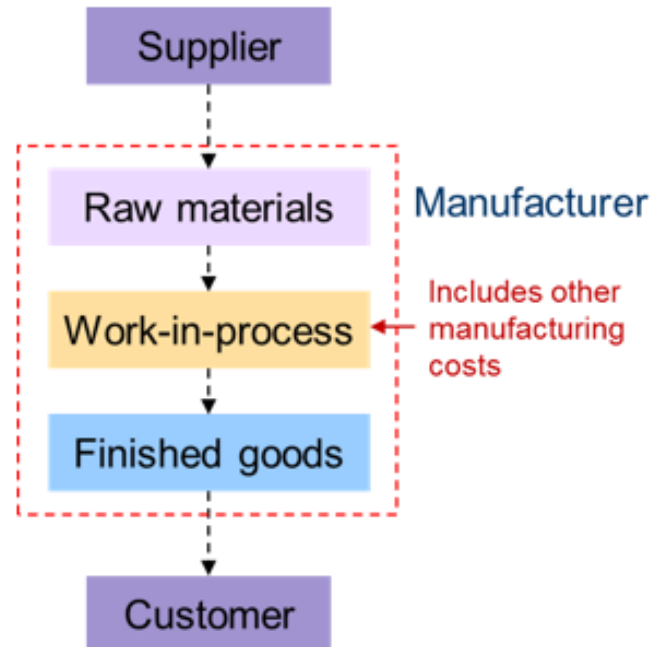
- Raw Materials – Basic parts not yet in production
- Working in Process – In production, not complete
- Finished Goods – Completed, ready for sale

Inventory Classification – Continued

Wholesaler or retailer:



Manufacturer:



Merchandising Company



Walmart Inc. Balance Sheets

Jan. 31, 2020

Current assets (in millions)

Cash and cash equivalents	\$ 9,465
Receivables, net	6,284
Inventories:	44,435
Prepaid expenses and other	1,622
Total current assets	<u>\$61,806</u>

Manufacturing Company



Procter & Gamble Company Balance Sheets

Jan. 30, 2020

Current assets (in millions)

Cash and cash equivalents	\$16,181
Accounts receivable	4,178
Inventories:	
Materials and supplies	\$1,414
Work in process	674
Finished goods	3,410
Total inventories	<u>5,498</u>
Prepaid expenses and other current assets	2,130
Total current assets	<u>\$27,987</u>

Recording Inventory Transactions

Recording Purchase of Inventory

At the time of Purchase – Company purchases \$500 of goods on credit.

Inventory (Dr.)	↑	\$500
A/P (Cr.)	↑	\$500

The cost is capitalized into the **Inventory asset** account immediately, not expensed.

Recording the Sale (Perpetual System) – Two Steps

A sale requires **two** distinct journal entries: one for revenue, one for cost.

Example: Company sold goods for \$1,000 on credit. Original cost was \$600.

Step 1: Recognize Revenue

Account Receivables (Dr.)	↑	\$800	
Sales Revenue (Cr.)	↑		\$800

Step 2: Match Cost (Expense)

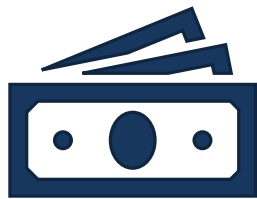
Cost of Goods Sold (Dr.)	↑	\$600	
Inventory (Cr.)	↑		\$600

Product vs. Period Costs

The Cost Principle: Product vs. Period Costs

When a cost is incurred, THE KEY question is:

- Did this expenditure help get the product ready for resale? Yes or No?



YES



Product Cost (Asset)

Stored on Balance Sheet

No



Period Cost (Expense)

Expense on Income Statement

Product Costs

All costs incurred to acquire goods and prepare them for sale (or in a sellable condition).



Merchandising

- Purchase Price
- Freight-in (Shipping to warehouse)
- Unpacking & Unloading
- Insurance in Transit

Manufacturing

- Direct materials
- Direct Labor
- Manufacturing Overhead

These costs are capitalized, and they live in the inventory account until sold.

Period Costs

Costs indirectly related to acquisition:

- Costs that are attributable to the passage of time rather than to the production of inventory.
- Costs incurred after the inventory has reached a saleable condition.



Key Examples

- **Selling Expenses:** Marketing, sales commissions, delivery to customers.
- **Administrative Expenses:** Executive salaries, office supplies.
- **Interest Costs:** Money paid to finance inventory purchases.

These costs are Expensed as Incurred.

They impact the Income Statement immediately, bypassing the Balance Sheet.

Product vs. Period Costs Visualization

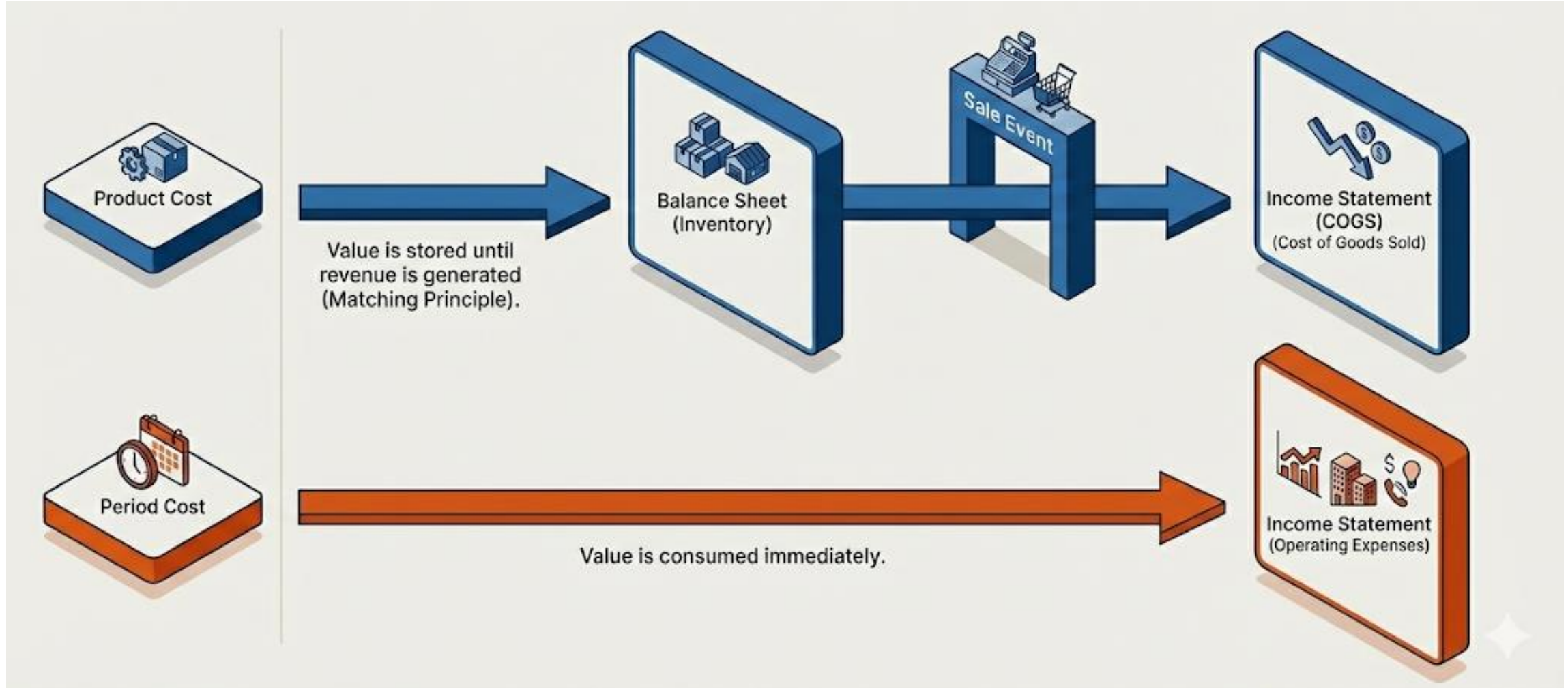
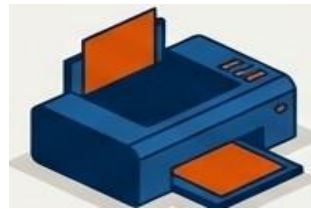


Illustration : The Office Depot Printer

Phase 1: Acquisition



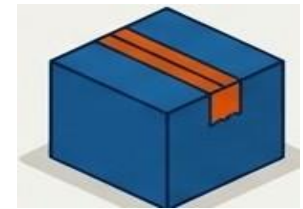
\$500 Cost

+



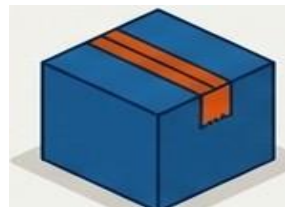
\$100 Freight-In
\$50 Insurance

=



Inventory Value: \$650

Phase 2: The Sale



COGS



\$50 Freight-Out

The \$50 delivery fee is recorded immediately as a selling expense.

Perpetual vs. Periodic Inventory System

The Tracking Systems: Periodic vs. Perpetual

$\text{Beginning Inventory} + \text{Net Purchases} = \text{COGS} + \text{Ending Inventory}$

Perpetual System (The Standard)



- Updates continuously at point of sale
- Ending Inventory derived in real-time
- Superior control and reporting

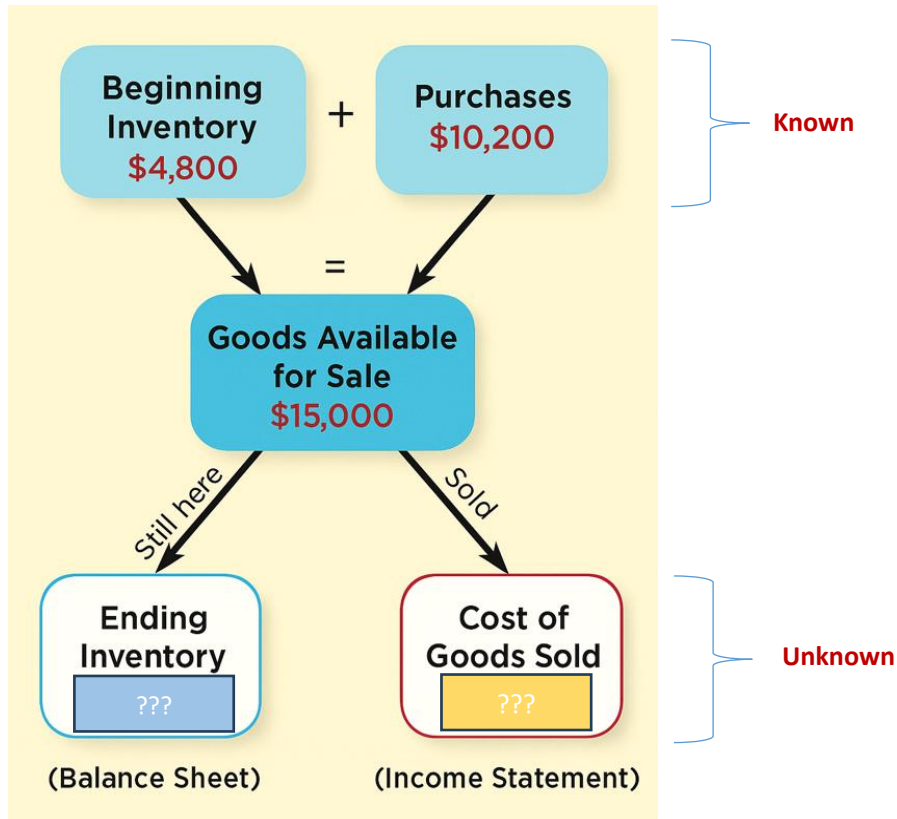
Periodic System (The Old Way)



- Updates only at end of period
- COGS unknown until physical count
- Formula: $\text{Beg Inv} + \text{Purchase} - \text{Ending Count} = \text{COGS}$

Example – Determining Cost of Goods Sold

Beginning Inventory+ Net Purchases= COGS+ Ending Inventory



Perpetual System (The Standard)

Beg Inv	+ Purchases	– COGS	= End Inv
Known	Known	Recorded	Recorded/Derived
\$4,800	\$10,200	\$9,000	\$6,000

Periodic System (The Old Way)

Beg Inv	+ Purchases	– End Inv	COGS
Known	Known	Counting	Recorded/Derived
\$4,800	\$10,200	\$6,000	\$9,000

Determining Ownership of Inventory

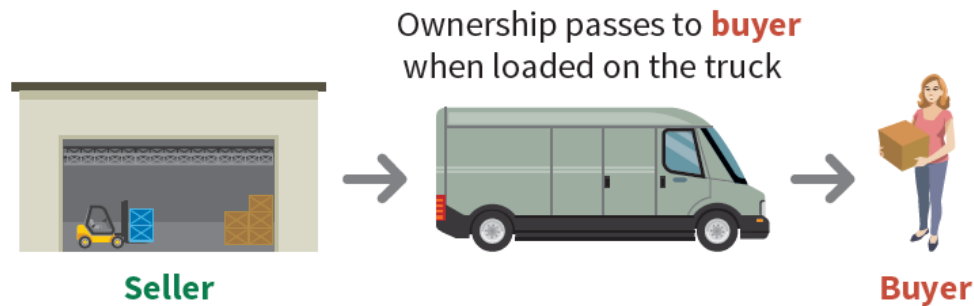
Inventory Ownership

Two questions related to the inventory ownership:

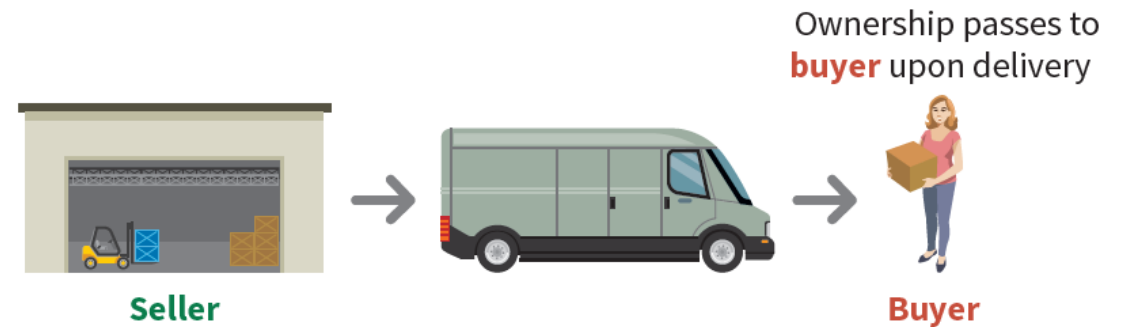
- (1) Do all the goods included in the physical count belong to the company?
- (2) Does the company own any goods that were not included in the count?

Inventory belongs to the company that holds legal title.

FOB Shipping Point: Buyer Pays Freight Costs



FOB Destination: Seller Pays Freight Costs



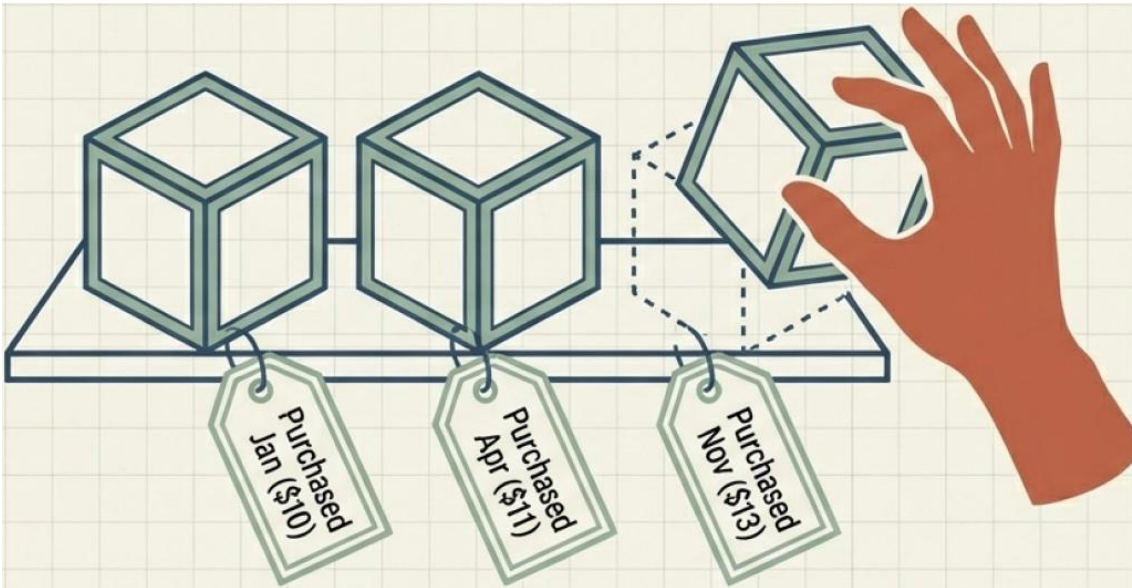
Consigned Goods: Title remains with the consignor, not the holder. Do not count them in consignee's inventory.

Costing Methods in Determining COGS and End Inv

Cost Flow Assumptions to Determine COGS

Dilemma

When Identical items are purchased at different times and different prices, which cost do we match to the sale?



Sold for \$20
What is the profit?

$$\$20 - \$10 = \$10?$$

$$\$20 - \$11 = \$9?$$

$$\$20 - \$13 = \$7?$$

Physical Flow $\neq$ Cost Flow : 1) Specific ID; 2) FIFO; 3) LIFO; 4) Average Cost Methods

Cost Flow Assumption 1 : First-in First-Out (FIFO)

- Suppose you open a bicycle store on 11/1. Your beginning inventory is zero. During November and December, you make the following bicycle purchases:
- During November and December, **you sold 7 bicycles.**
- Compute COGS and Ending Inventory

<u>Purchases</u>	<u>Quantity</u>	<u>Unit Cost</u>	<u>Total Cost</u>
11/2	5	\$100	\$500
12/8	4	\$125	\$500
12/20	6	\$130	\$780
Goods Available for Sale	15		\$1,780

FIFO: COGS (7 bikes sold)

$$\begin{array}{rcl} 5 @ \$100 & = & \$500 \\ \underline{2 @ \$125} & = & \underline{\$250} \\ 7 & = & \$750 \text{ (COGS)} \end{array}$$

FIFO: Ending Inventory (8 bikes remaining)

$$\begin{array}{rcl} 6 @ \$130 & = & \$500 \\ \underline{2 @ \$125} & = & \underline{\$250} \\ 8 & = & \$1,030 \text{ (EI)} \end{array}$$

Cost Flow Assumption 2 : Last-In Last-Out(LIFO)

- Suppose you open a bicycle store on 11/1. Your beginning inventory is zero. During November and December, you make the following bicycle purchases:
- During November and December, **you sold 7 bicycles.**
- Compute COGS and Ending Inventory

<u>Purchases</u>	<u>Quantity</u>	<u>Unit Cost</u>	<u>Total Cost</u>
11/2	5	\$100	\$500
12/8	4	\$125	\$500
12/20	6	\$130	\$780
Goods Available for Sale	15		\$1,780

LIFO: COGS (7 bikes sold)

$$\begin{array}{rcl} 6 @ \$130 & = & \$780 \\ 1 @ \$125 & = & \$125 \\ \hline 7 & = & \$905 \text{ (COGS)} \end{array}$$

LIFO: Ending Inventory (8 bikes remaining)

$$\begin{array}{rcl} 5 @ \$100 & = & \$500 \\ 3 @ \$125 & = & \$375 \\ \hline 8 & = & \$875 \text{ (EI)} \end{array}$$

Cost Flow Assumption 3 : Average Cost

- Suppose you open a bicycle store on 11/1. Your beginning inventory is zero. During November and December, you make the following bicycle purchases:
- During November and December, **you sold 7 bicycles.**
- Compute COGS and Ending Inventory

<u>Purchases</u>	<u>Quantity</u>	<u>Unit Cost</u>	<u>Total Cost</u>
11/2	5	\$100	\$500
12/8	4	\$125	\$500
12/20	6	\$130	\$780
Goods Available for Sale	15		\$1,780

The average cost of bike = Goods available for sale / # of bikes available for sale
= \$1,780 / 15 = \$118.67 / bike

Avg Cost: COGS (7 bikes sold)

7 @ 118.67 = \$831

Avg Cost: Ending Inventory (8 bikes remaining)

8 x \$118.67 = \$949

The Effect of Cost Flow Assumptions on F/S

In period of rising prices, LIFO reports the lowest net income, resulting in the lowest tax liability.

		FIFO	Average	LIFO
	Sales	\$20,000	\$20,000	\$20,000
Lowest	Cost of Goods Sold	7,500	8,307	9,050
	Gross Profit	12,500	11,693	10,950
	Admin & Selling Expenses	7,700	7,700	7,700
	Income before Taxes	4,800	3,993	3,250
	Income Tax Expense	960	800	650
Highest	Net Income	\$3,840	\$3,193	\$2,600
	Inventory Balance at Year-End	\$10,300	\$9,493	\$8,750

Inventory Valuation after Purchase

The Problem: Obsolete Inventory

Case Study: MTP, Inc.

MTP, Inc. has been selling typewriters and fax machines for 30 years. Due to technological advancement, these products are becoming obsolete in the late 1990's. Inventory recorded at: **\$300,000** & Actual market value: **\$100,000**



Obsolete



Obsolete

? What should the accountant report on the balance sheet?

Answer: Report inventory at \$100,000 and recognize a \$200,000 loss immediately.

Journal Entry:

Dr. Loss on Inventory Write-down	\$200,000 ↑
Cr. Inventory	\$200,000 ↓

Accounting Conservatism

Principle: Recognize losses immediately, defer gains until realized



LOSSES

Recognize IMMEDIATELY
when they become known

- Write down inventory now
- Record loss on income statement



GAINS

DEFER until realized
(actually sold)

- Do NOT write up inventory
- Wait until sale occurs

Why? To prevent overstating assets and income — protecting investors and creditors.

What if Market Value Increases? – IFRS vs. U.S. GAAP

Scenario: Typewriters suddenly become popular! Market value jumps to \$400,000.



IFRS

REVERSAL ALLOWED

- Can write back UP to \$300,000
- Limited to ORIGINAL cost only
- Cannot exceed \$300,000
- Recognize \$200,000 gain



U.S. GAAP

DO NOTHING

- Inventory stays at \$100,000
- Write-down is PERMANENT
- No reversal allowed
- Gain recognized only when sold

The LCNRV Rule (**L**ower-of-**C**ost or **N**et **R**ealizable **V**alue)

Report inventory at the LOWER of:

1. **Cost** — what you originally paid
2. **Net Realizable Value (NRV)** — expected selling price minus costs to complete and sell

NRV Formula

NRV = Selling Price – Cost to Complete – Selling Costs

(What you'll actually receive in your pocket)

Decision Process:

Cost

vs

NRV



Report the LOWER

LCNRV Example : Starbucks

Given: Unfinished inventory

Inventory Cost:	\$950
Estimated Selling Price:	\$1,000
Cost to Complete:	\$50
Selling Costs:	\$200

NRV Calculation

Selling Price	\$1,000
– Cost to Complete	(\$50)
– Selling Costs	(\$200)
= NRV	\$750

Compare:

Cost
\$950

>

NRV
\$750 ✓

Report at
\$750

Journal Entry:

Dr. Loss Due to Decline in Inventory	\$200 ↑
Cr. Inventory	\$200 ↓

Concept Check

If NRV rises after an inventory write-down, how does IFRS differ from U.S. GAAP?

- A. IFRS also prohibits reversals
- B. IFRS allows reversals up to the original cost**
- C. IFRS allows unlimited upward revaluations
- D. IFRS requires immediate recognition of any gain above cost

Inventory Analysis Ratio

Inventory Management Ratios

Inventory Turnover Ratio

$$\frac{\text{COGS}}{\text{Average Inventory}}$$

How many times inventory is sold during the period

$$\text{Avg Inventory} = (\text{Beginning} + \text{Ending}) \div 2$$

Days in Inventory

$$\frac{365}{\text{Inventory Turnover}}$$

Average days to sell inventory (also = Avg. Inventory age)

Example Calculation

COGS	\$80,000
Beginning Inventory	\$14,000
Ending Inventory	\$18,000

$$\text{Avg Inv} = (\$14\text{K} + \$18\text{K}) \div 2 = \$16,000$$

Turnover Ratio
5 times

Days in Inventory
73 days

Turnover = $\$80,000 / \$16,000 = 5 \text{ times/year}$
Days = $365 / 5 = 73 \text{ days}$

Interpreting the Ratios

HIGH Turnover

(Low Days in Inventory)

- Efficient inventory management
- Strong sales performance
- Lower holding costs
- Risk: Stockouts & lost sales

LOW Turnover

(High Days in Inventory)

- Too much inventory on hand
- Higher storage & insurance costs
- Risk of obsolescence/damage
- Cash tied up in inventory

Goal: Find the RIGHT balance

Enough inventory to meet demand, but not so much that costs pile up.